



Bank Customer Churn Analysis

Data Analysis Using Python

Technologies Used : Python

Libraries Used : Pandas • NumPy • Matplotlib • Seaborn • Plotly • Streamlit

Development Environment: Jupyter Notebook

Dashboard : Streamlit

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INTRODUCTION



Customer churn is one of the biggest challenges faced by banks. Identifying customers who are likely to leave enables organizations to take proactive retention measures.



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ABOUT THE DATA



Dataset Name : Bank Customer Churn Dataset



Source : Kaggle



Total Records : 10,000



Total Features : 14



Target Variable : Exited (0-Retained , 1- Churned)



Feature Include : Credit score , Geography, Gender, Age, Tenure, Balance, etc.





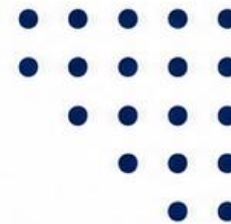
3 PROBLEM STATEMENT

The bank is facing a high customer churn rate.

We need to identify the key factors contributing to customer churn and understand patterns in customer behavior.

This analysis will help the bank design strategies to retain existing customers and reduce churn





4 DATA INSIGHTS



Total Customers

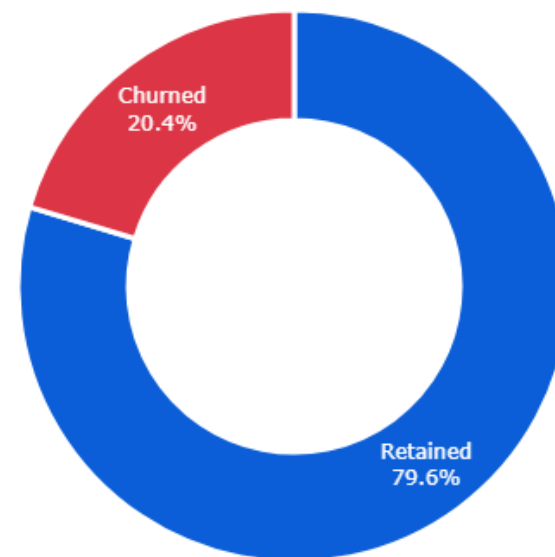
10,000



Churned Customer

2,037

Customer Churn Distribution



Customer Status

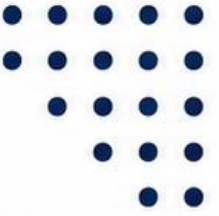
■ Retained

■ Churned

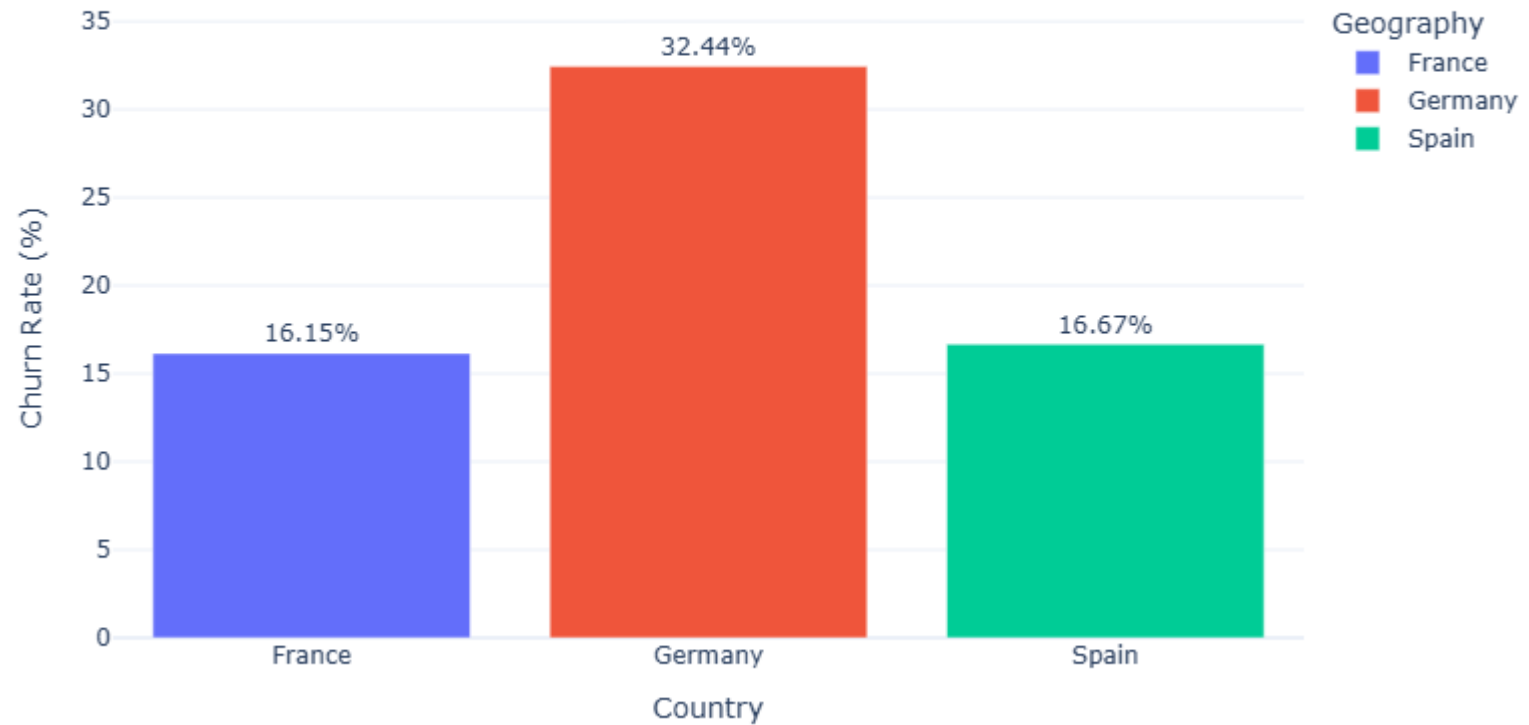
Overall Churn Rate is **20.4 %**

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DATA INSIGHTS

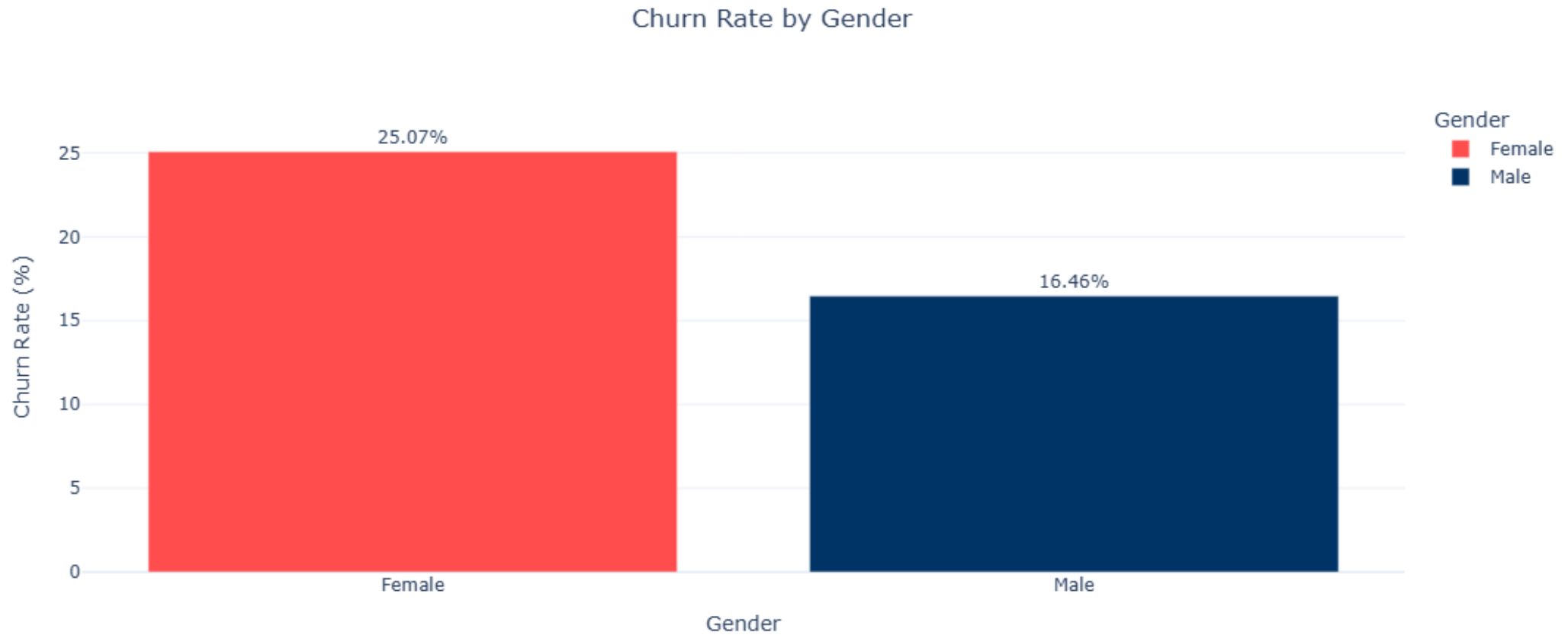
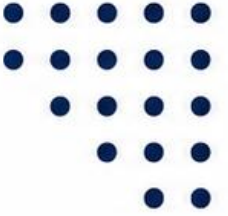


Churn Rate by Geography



Geography: Germany has the highest churn rate among all regions

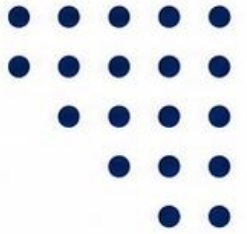
6 DATA INSIGHTS



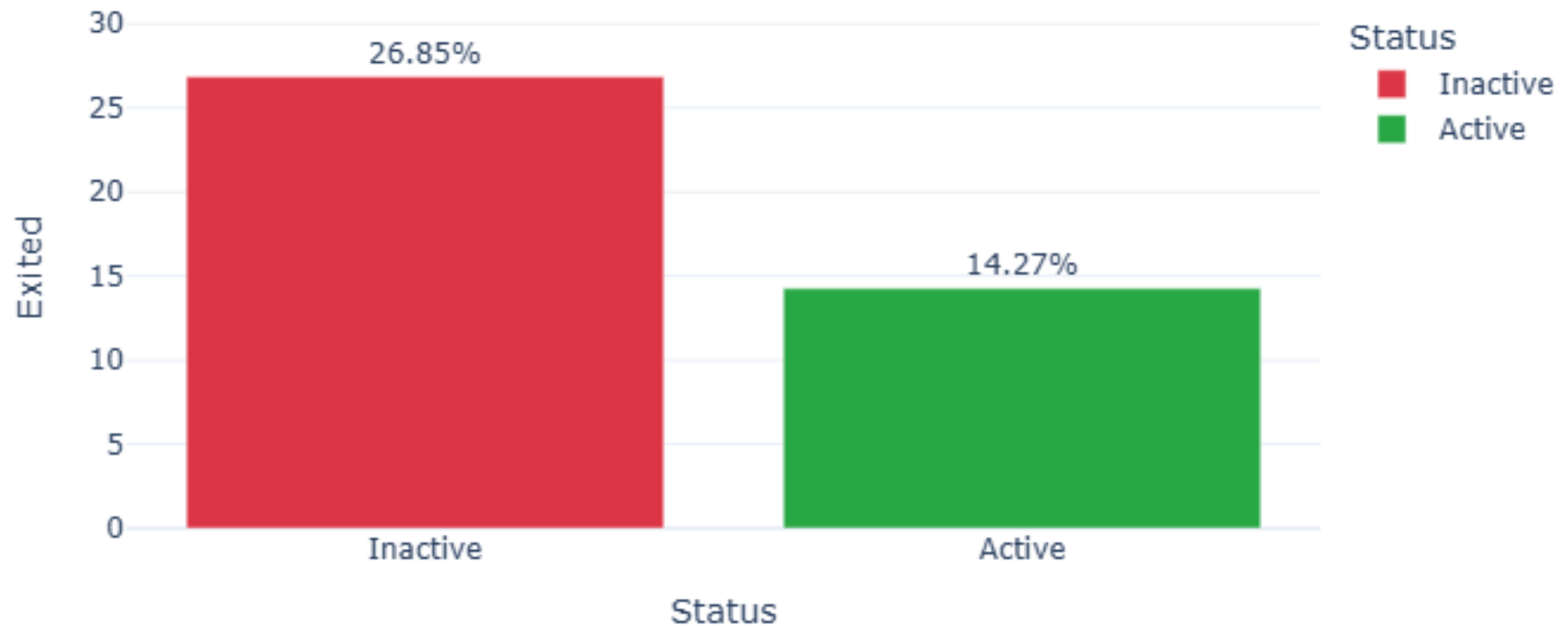
Female customers show a higher churn rate than male customers.

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DATA INSIGHTS



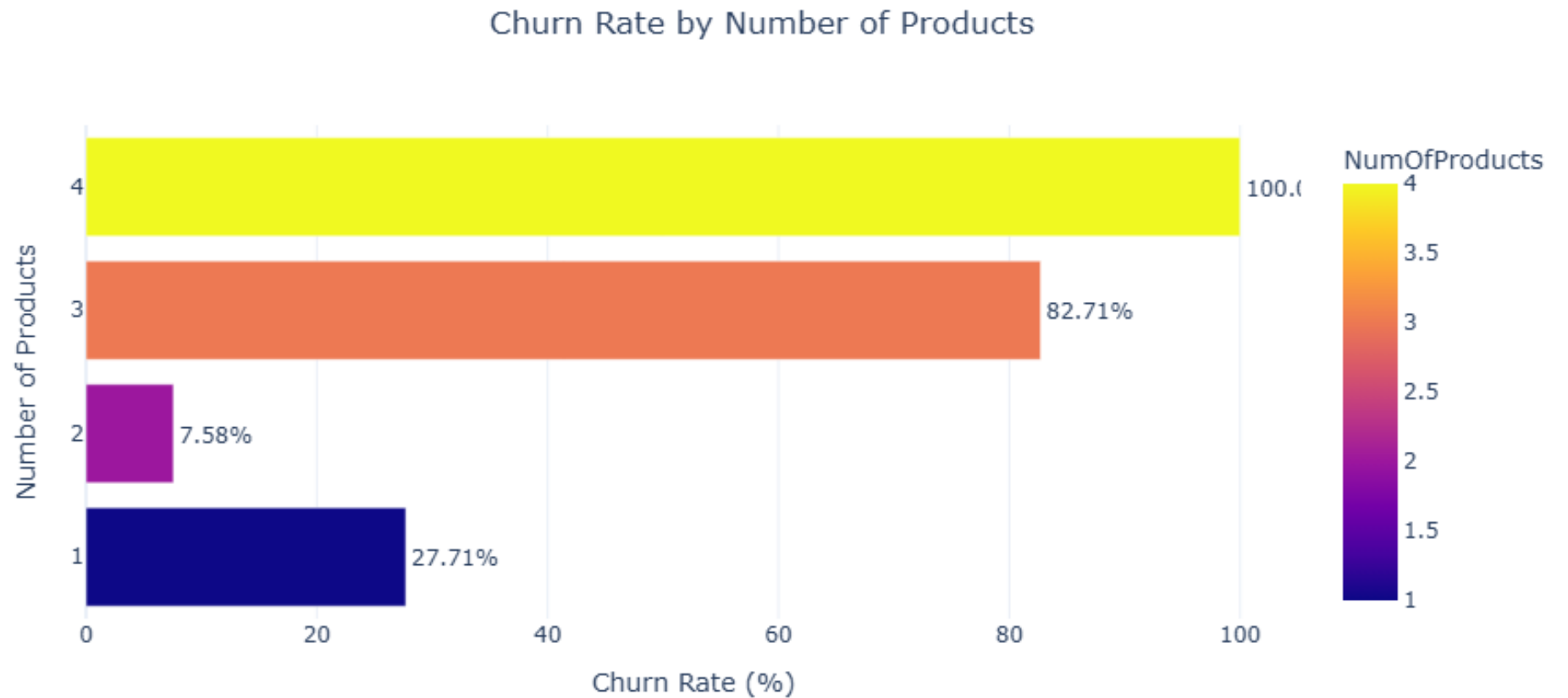
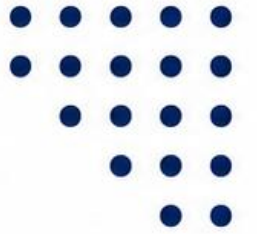
Churn Rate by Active Member



Inactive customers exhibit nearly twice the churn rate of active customers.

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DATA INSIGHTS



Customers with fewer products show significantly higher churn rates.

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PROPOSED SOLUTION



Personalized Engagement

Engage inactive customer with personalized offers and communication.



Loyalty Programs

Introduce loyalty rewards for long term customers.



Cross Selling Opportunities

Encourage customers to use more products and services .



Proactive Customer Support

Identify at risk customers early and provide proactive support .



10 CONCLUSION



Overall Churn Rate is **20.4 %**.



Germany has the highest churn rate.



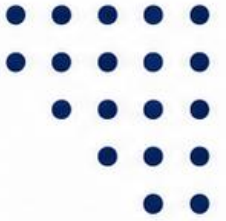
Inactive customer churn more.



Customer with fewer products are more likely to leave.



Data driven strategies can help reduce churn and improve retention.



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FUTURE SCOPE



Predictive Modeling

Develop machine learning models to predict customer churn with higher accuracy..



Real Time Dashboard

Integrate live data sources for continuous monitoring and decision-making.



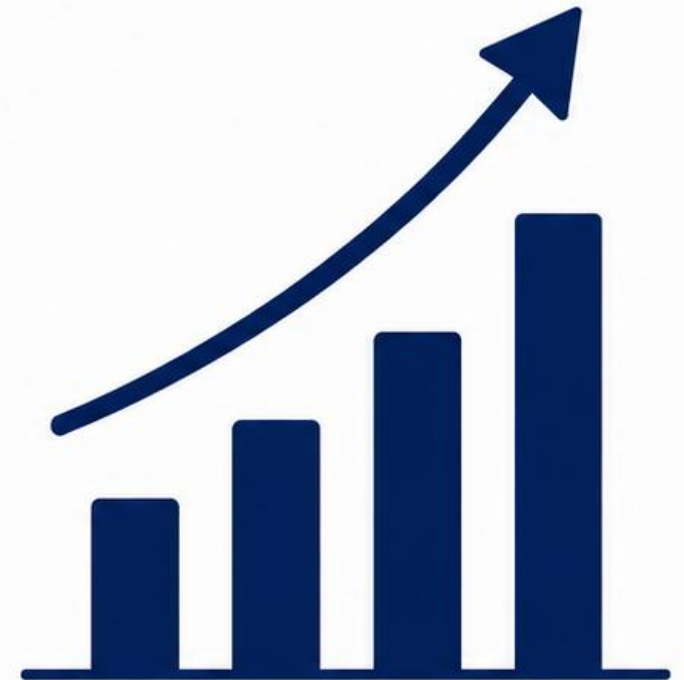
Customer Segmentation

Build targeted retention strategies based on customer behavior and demographics.



Advanced Analytics

Leverage AI and advanced analytics to generate deeper business insights.





THANK YOU!

